

AZ-305

Azure Solutions Architect Interview Preparation Guide

300 Curated Q&A · 7 Core Domains · All Exam Topics Covered

Beginner · Intermediate · Advanced

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Introduction

About This Guide

This guide is a comprehensive interview preparation resource for the **Microsoft Azure Solutions Architect Expert (AZ-305)** certification and related job interviews. It covers all major exam domains with 300 carefully curated questions and answers, ranging from foundational concepts to advanced architectural decisions.

Who Is This For?

Designed for cloud architects, senior engineers, and IT professionals preparing for AZ-305 certification interviews, technical architecture discussions, or senior cloud engineering roles. Whether transitioning from on-premises to cloud or advancing existing Azure expertise, this guide provides structured revision across all key topics.

How to Use This Guide

Work through each section systematically or jump to areas needing the most preparation. **Bolded keywords** highlight critical terminology. Focus on understanding the reasoning behind architectural choices — interviewers often ask 'why' as much as 'what'. Use this guide alongside hands-on Azure portal practice for best results.

Exam Domain Coverage

The AZ-305 exam covers: Design Identity, Governance and Monitoring Solutions (25-30%), Design Data Storage Solutions (15-20%), Design Business Continuity Solutions (10-15%), and Design Infrastructure Solutions (30-35%). This guide maps its sections to these domains for targeted study.

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Section 1: Azure Architecture Design

Azure architecture design forms the backbone of the AZ-305 exam. Questions test your ability to design scalable, resilient, and cost-effective solutions using Azure Well-Architected Framework principles, landing zones, and hybrid architectures.

Q1.

What is the Azure Well-Architected Framework?

Answer:

The Azure Well-Architected Framework is a set of guiding principles to improve the quality of a workload. It consists of five pillars: **Reliability**, **Security**, **Cost Optimization**, **Operational Excellence**, and **Performance Efficiency**. Each pillar provides best practices and design considerations for building robust Azure solutions.

Q2.

What is an Azure Landing Zone?

Answer:

An Azure Landing Zone is a pre-configured, scalable environment that provides a foundation for hosting workloads in Azure. It encompasses identity, governance, networking, and security controls aligned to enterprise standards, enabling teams to quickly deploy workloads in a secure and compliant manner.

Q3.

Explain the difference between scale-up and scale-out in Azure.

Answer:

Scale-up (vertical scaling) means increasing the size of an existing resource (e.g., VM size). **Scale-out (horizontal scaling)** means adding more instances of a resource. Scale-out is preferred for cloud-native workloads as it provides higher availability and elasticity without downtime.

Q4.

What is Azure Advisor?

Answer:

Azure Advisor is a personalized cloud consultant that analyzes your Azure usage and provides recommendations across **Cost**, **High Availability**, **Security**, **Performance**, and **Operational Excellence**. It helps optimize Azure resources based on actual telemetry and best practices.

Q5.

What is the purpose of Azure Resource Manager (ARM)?

Answer:

ARM is the deployment and management service for Azure. It provides a unified layer that handles authentication, authorization, and resource lifecycle management. ARM enables **Infrastructure as Code (IaC)** using ARM templates or Bicep, consistent resource tagging, and role-based access control across all Azure resources.

Q6.

How do you design for high availability in Azure?

Answer:

High availability in Azure is achieved through: **Availability Sets** (protecting VMs from hardware failures), **Availability Zones** (isolated datacenters within a region), **Azure Load Balancer** for traffic distribution, **Traffic Manager** for DNS-level failover, and **geo-redundant storage and databases**. SLA targets dictate the combination of these features.

Q7.

What is the difference between an Availability Set and an Availability Zone?

Answer:

An **Availability Set** protects against in-rack hardware failures within a single datacenter using fault domains and update domains, providing 99.95% SLA. An **Availability Zone** protects against entire datacenter failures by distributing VMs across physically separate zones within a region, providing 99.99% SLA.

Q8.

What is Azure Service Health?

Answer:

Azure Service Health provides personalized alerts and guidance when Azure service issues, planned maintenance, or health advisories affect your resources. It includes **Azure Status** (global outages), **Service Health** (regional issues), and **Resource Health** (individual resource status).

Q9.

What is the Azure Architecture Center?

Answer:

The Azure Architecture Center is Microsoft's online repository of architectural guidance, reference architectures, design patterns, and best practices. It helps architects design solutions using proven patterns for common scenarios like microservices, data pipelines, IoT, and enterprise integration.

Q10.

How do you approach multi-region architecture in Azure?

Answer:

Multi-region architecture uses **active-active** or **active-passive** patterns. Key components include Azure Front Door or Traffic Manager for global load balancing, geo-replicated databases (Cosmos DB, SQL geo-replication), Azure CDN for static content, and paired regions for data residency and compliance. RTO and RPO dictate the design.

Q11.

What is Azure Front Door and when would you use it?

Answer:

Azure Front Door is a global, scalable entry point for fast delivery of web applications. It provides **global HTTP load balancing**, SSL offloading, URL-based routing, WAF integration, and anycast routing for low latency. Use it when you need global reach, fast failover, and application-layer (Layer 7) routing.

Q12.

Explain Azure regions and region pairs.

Answer:

An Azure **region** is a geographic area containing one or more datacenters. Azure **region pairs** are two regions within the same geography that Microsoft designates for cross-region replication. In the event of an outage, services can fail over to the paired region. Updates are also rolled out sequentially across pairs to reduce risk.

Q13.

What is Azure Blueprints?

Answer:

Azure Blueprints enables cloud architects to define a repeatable set of Azure resources that implement organizational standards. A blueprint can include **role assignments**, **policy assignments**, **ARM templates**, and **resource groups**. Unlike ARM templates alone, Blueprints track deployed resources and maintain the relationship over time.

Q14.

How does Azure Policy differ from Azure RBAC?

Answer:

Azure Policy enforces organizational rules on resource properties (e.g., allowed VM sizes, required tags, enforced encryption) and evaluates compliance. **Azure RBAC** controls who can perform actions on resources (e.g., who can create or delete VMs). RBAC handles authorization while Policy handles governance and compliance.

Q15.

What is Azure Migrate?

Answer:

Azure Migrate is a hub of tools for assessing and migrating on-premises workloads to Azure. It includes assessment tools for servers, databases, and web apps; dependency visualization; TCO calculator integration; and migration tools supporting **lift-and-shift**, **re-platform**, and **re-architect** migration strategies.

Section 2: Identity and Governance

Identity and governance are central to enterprise Azure deployments. This section covers Azure Active Directory, role-based access control, privileged identity management, and compliance frameworks essential for the AZ-305 exam.

Q16.

What is Azure Active Directory (Azure AD)?

Answer:

Azure AD is Microsoft's cloud-based identity and access management service. It provides **single sign-on (SSO)**, **multi-factor authentication (MFA)**, **conditional access**, and **B2B/B2C collaboration**. It supports modern authentication protocols like OAuth 2.0, OpenID Connect, and SAML 2.0.

Q17.

What is Azure AD B2C?

Answer:

Azure AD B2C (Business-to-Consumer) is a customer identity and access management (CIAM) solution. It enables organizations to build custom authentication flows for consumer-facing applications, supporting social identity providers (Google, Facebook) and custom user journeys via user flows or custom policies.

Q18.

Explain Conditional Access in Azure AD.

Answer:

Conditional Access is Azure AD's policy engine that evaluates signals (user, location, device, app, risk) to enforce access decisions. Policies can **block access**, **require MFA**, or **require compliant devices**. It implements Zero Trust principles by granting least-privilege access based on contextual conditions.

Q19.

What is Privileged Identity Management (PIM)?

Answer:

Azure AD PIM provides **just-in-time privileged access** to Azure AD and Azure resources. Users request elevated roles for a limited time, require justification and approval, and all activations are logged. PIM reduces the attack surface by eliminating permanent admin role assignments.

Q20.

What is the difference between Azure AD roles and Azure RBAC roles?

Answer:

Azure AD roles manage resources within Azure AD (e.g., Global Administrator, User Administrator). **Azure RBAC roles** manage access to Azure resources like VMs, storage, and networking (e.g., Owner, Contributor, Reader). Some roles span both, such as the Global Administrator who can manage Azure subscriptions.

Q21.

What is Azure AD Identity Protection?

Answer:

Identity Protection uses machine learning to detect **sign-in risks** (e.g., anonymous IP, impossible travel) and **user risks** (e.g., leaked credentials). It can automatically block or require MFA for risky sign-ins and integrates with Conditional Access policies for automated remediation.

Q22.

What is a Managed Identity in Azure?

Answer:

A Managed Identity provides Azure services with an automatically managed identity in Azure AD. There are two types: **System-assigned** (tied to a specific resource's lifecycle) and **User-assigned** (standalone identity shared across multiple resources). It eliminates the need to store credentials in code.

Q23.

Explain Azure AD Connect.

Answer:

Azure AD Connect synchronizes on-premises Active Directory identities to Azure AD, enabling hybrid identity. It supports **password hash sync**, **pass-through authentication**, and **federation with AD FS**. Azure AD Connect Health provides monitoring of the synchronization infrastructure.

Q24.

What is Azure AD External Identities?

Answer:

Azure AD External Identities enables secure collaboration with partners and customers outside your organization. **B2B collaboration** lets external users sign in with their existing identities. **B2B direct connect** enables shared Teams channels. It supports MFA, Conditional Access, and entitlement management for guest users.

Q25.

What is entitlement management in Azure AD?

Answer:

Entitlement management is an identity governance feature that automates access request workflows, assignments, reviews, and expiration for **access packages**. It enables organizations to manage access to groups, applications, and SharePoint sites at scale, especially for external partners.

Q26.

What is Microsoft Defender for Identity?

Answer:

Microsoft Defender for Identity monitors on-premises Active Directory signals to detect, identify, and investigate advanced threats, compromised identities, and insider attacks. It detects techniques like **pass-the-hash**, **Kerberoasting**, and **lateral movement**.

Q27.

How do you implement least privilege access in Azure?

Answer:

Implement least privilege by: using **Azure RBAC** scoped to the lowest necessary level, enabling **PIM** for just-in-time admin access, applying **Conditional Access** policies, conducting regular **access reviews**, and removing unused permissions and service principal secrets.

Q28.

What is Azure AD Application Proxy?

Answer:

Azure AD Application Proxy enables secure remote access to on-premises web applications without a VPN or opening inbound firewall ports. A lightweight connector establishes outbound connections to Azure, providing SSO, Conditional Access, and MFA for legacy apps.

Q29.

What is Microsoft Entra?

Answer:

Microsoft Entra is a family of identity and access products from Microsoft. It includes **Azure AD** (Microsoft Entra ID), **Entra Permissions Management** (CIEM), **Entra Verified ID** (decentralized identity), and **Entra Workload ID** (for apps and services).

Q30.

What is Azure Policy Initiative?

Answer:

A Policy Initiative (Policy Set) is a collection of Azure Policy definitions grouped to achieve a single governance goal, such as the **CIS Microsoft Azure Foundations Benchmark**. Initiatives simplify assignment and compliance reporting across multiple policies at once.

Section 3: Data Storage Solutions

Choosing the right data storage solution is a critical architectural decision. This section explores Azure's diverse storage options – from blob and file storage to relational and NoSQL databases – along with data migration and integration strategies.

Q31.

What are the main Azure storage account types?

Answer:

Azure offers **Standard General-Purpose v2** (most recommended, supports all storage services), **Premium Block Blobs** (low-latency blob workloads), **Premium File Shares** (high-performance file shares), and **Premium Page Blobs** (unmanaged VM disks). The account type determines performance tier and supported services.

Q32.

Explain Azure Blob Storage access tiers.

Answer:

Azure Blob Storage has four access tiers: **Hot** (frequent access, higher storage cost), **Cool** (infrequent access, min 30-day retention), **Cold** (rarely accessed, min 90-day retention), and **Archive** (offline, lowest cost, min 180-day retention, hours to rehydrate). Lifecycle management policies automate tier transitions.

Q33.

What is Azure Data Lake Storage Gen2?

Answer:

ADLS Gen2 is built on Azure Blob Storage with a **hierarchical namespace** enabled, providing file system semantics, directory-level ACLs, and high performance for analytics workloads. It integrates natively with Azure Synapse Analytics, Azure Databricks, and HDInsight for big data processing.

Q34.

When would you choose Azure Cosmos DB over Azure SQL Database?

Answer:

Choose **Cosmos DB** for globally distributed, multi-model, schema-flexible workloads requiring single-digit millisecond latency, multi-region writes, and automatic horizontal scaling. Choose **Azure SQL Database** for relational, ACID-compliant workloads with complex queries and structured schemas.

Q35.

What is Azure SQL Managed Instance?

Answer:

Azure SQL Managed Instance is a fully managed PaaS with near 100% SQL Server compatibility, including SQL Agent, linked servers, CLR, and cross-database queries. Ideal for **lift-and-shift migrations** from on-premises SQL Server. It runs in a dedicated VNet providing network isolation.

Q36.

Explain the Cosmos DB consistency levels.

Answer:

Cosmos DB offers five consistency levels (strongest to weakest): **Strong**, **Bounded Staleness**, **Session** (default), **Consistent Prefix**, and **Eventual**. Stronger consistency guarantees accuracy but increases latency; weaker levels improve performance and availability.

Q37.

What is Azure Table Storage and when would you use it?

Answer:

Azure Table Storage is a NoSQL key-value store for structured, non-relational data. It is cost-effective for storing large volumes of structured data. Use it for **web scale datasets**, **device data**, or **user data** when you need flexible schemas without complex joins or foreign keys.

Q38.

What is the difference between Azure Files and Azure Blob Storage?

Answer:

Azure Files provides fully managed file shares accessible via SMB and NFS protocols, making it a drop-in replacement for on-premises file servers. **Azure Blob Storage** is an object store optimized for unstructured data accessed via HTTP endpoints. Files supports persistent connections; Blobs are accessed by URL.

Q39.

What is Azure Database for PostgreSQL Flexible Server?

Answer:

A fully managed PostgreSQL service offering **zone-redundant high availability**, **burstable compute**, automatic backups, and point-in-time restore. It supports scheduled maintenance windows and stop/start to reduce costs, suitable for production PostgreSQL workloads.

Q40.

How do you secure data at rest in Azure Storage?

Answer:

Azure Storage encrypts all data at rest using **Storage Service Encryption (SSE)** with 256-bit AES. Customers can use **Microsoft-managed keys**, **customer-managed keys (CMK)** in Azure Key Vault, or **customer-provided keys**. Azure Disk Encryption uses BitLocker (Windows) or dm-crypt (Linux) for VM disks.

Q41.

What is Azure Synapse Analytics?

Answer:

Azure Synapse Analytics is an enterprise analytics service combining **data warehousing**, **big data analytics**, and **data integration**. It includes Synapse SQL (serverless and dedicated pools), Synapse Spark, Synapse Pipelines, and integrates with Power BI and Azure Machine Learning for end-to-end analytics.

Q42.

What is Azure Cache for Redis?

Answer:

Azure Cache for Redis is a fully managed, in-memory data store for **caching**, **session management**, **message brokering**, and **leaderboards**. It supports geo-replication, persistence, clustering, and VNet integration for enterprise scenarios requiring low-latency data access.

Q43.

When would you use Azure Queue Storage vs Azure Service Bus?

Answer:

Use **Azure Queue Storage** for simple, large-volume message queuing with messages up to 64 KB. Use **Azure Service Bus** for enterprise messaging with **ordered delivery**, **dead-letter queues**, **duplicate detection**, **sessions**, and messages up to 100 MB.

Q44.

What is soft delete in Azure Storage?

Answer:

Soft delete retains deleted blobs, containers, or file shares for a configurable retention period (1-365 days). Deleted items can be recovered during the retention window. It protects against accidental deletion and ransomware attacks, enabled per storage account and works with blob versioning.

Q45.

What is Azure Data Factory?

Answer:

Azure Data Factory (ADF) is a cloud-based ETL and data integration service for creating, scheduling, and orchestrating **data pipelines** across 90+ connectors. ADF supports **data flows** for code-free transformations, SSIS package execution, and integration with Azure Synapse.

Section 4: Business Continuity

Business continuity and disaster recovery (BCDR) design is a key domain in AZ-305. Architects must understand RTO, RPO, backup strategies, failover mechanisms, and Azure-native BCDR services to ensure organizations meet their resilience and compliance requirements.

Q46.

What is the difference between RTO and RPO?

Answer:

RTO (Recovery Time Objective) is the maximum acceptable downtime after a disaster — how quickly systems must be restored. **RPO (Recovery Point Objective)** is the maximum acceptable data loss measured in time — the age of the recovery point. Lower RTO/RPO requires more expensive, active solutions like hot standby replicas.

Q47.

What is Azure Site Recovery (ASR)?

Answer:

Azure Site Recovery orchestrates replication, failover, and recovery of VMs and physical servers to Azure or a secondary site. It supports Hyper-V, VMware, and Azure VMs. ASR provides **continuous replication**, **one-click failover**, **automated recovery plans**, and **non-disruptive DR drills** to test failover without impacting production.

Q48.

What is Azure Backup?

Answer:

Azure Backup is a cloud-based backup service protecting Azure VMs, SQL databases, SAP HANA, Azure Files, and on-premises workloads. It stores backups in **Recovery Services Vaults** with geo-redundant storage, supports **soft delete**, **cross-region restore**, and provides centralized management via the Backup Center.

Q49.

Explain the difference between GRS and ZRS storage redundancy.

Answer:

GRS (Geo-Redundant Storage) replicates data synchronously three times locally and asynchronously to a paired region, protecting against regional outages. **ZRS (Zone-Redundant Storage)** replicates data synchronously across three availability zones within a region, protecting against zone-level failures with lower latency.

Q50.

What is a Recovery Services Vault?

Answer:

A Recovery Services Vault stores backup data and Site Recovery configurations. It provides **role-based access control**, **backup policy management**, **multi-source backup**, **monitoring and alerts**, and supports cross-region restore for Azure VM backups to paired regions.

Q51.

What storage redundancy options are available in Azure?

Answer:

Azure offers: **LRS** (3 copies in one datacenter), **ZRS** (3 copies across zones), **GRS** (LRS + async replication to paired region), **GZRS** (ZRS + async replication to paired region), and read-access variants **RA-GRS** and **RA-GZRS** enabling reads from the secondary region.

Q52.

How does Azure SQL Database handle high availability?

Answer:

Azure SQL Database uses an **Always On** based architecture. The General Purpose tier uses remote storage with a standby replica. The Business Critical tier maintains 3-4 local replicas using Always On AG, offering 99.99% SLA. **Auto-failover groups** enable transparent geo-failover across regions.

Q53.

What is an Azure auto-failover group?

Answer:

An auto-failover group manages geo-replication and failover of a group of databases to another region. It provides a **read-write listener** and **read-only listener** DNS endpoint, abstracting connection strings so applications automatically reconnect after failover without code changes.

Q54.

What is the Azure Backup soft delete feature?

Answer:

Soft delete in Azure Backup retains deleted backup data for 14 additional days, protecting against accidental or malicious deletion. The data can be recovered at no extra cost. Enhanced soft delete extends this to 180 days and can be made irreversible, providing ransomware protection.

Q55.

How do you design a disaster recovery strategy for Azure VMs?

Answer:

Design DR using: **Azure Site Recovery** for VM replication to a secondary region, **auto-failover groups** for databases, **Azure Backup** for point-in-time recovery, **Traffic Manager** or **Front Door** for DNS failover, and **infrastructure as code** to rapidly redeploy resources. Document and test the DR plan regularly.

Q56.

What is the Azure Business Continuity Center?

Answer:

Azure Business Continuity Center is a unified management experience for backup and disaster recovery across Azure and hybrid workloads. It provides a single dashboard to view protection status, configure policies, monitor jobs, and manage compliance for both Azure Backup and Azure Site Recovery.

Q57.

Explain Cosmos DB built-in high availability.

Answer:

Cosmos DB automatically replicates data across all configured regions. With **multi-region write** enabled, any region can accept writes with conflict resolution. Cosmos DB provides **99.999% SLA** for multi-region accounts and performs automatic failover if a region becomes unavailable.

Q58.

What is the difference between backup and replication?

Answer:

Backup creates periodic point-in-time copies stored separately, protecting against data corruption, accidental deletion, and ransomware. **Replication** continuously mirrors data to another location, protecting against infrastructure failures with near-zero RPO but no protection against data corruption events.

Q59.

What is Azure Elastic SAN?

Answer:

Azure Elastic SAN is a cloud-native Storage Area Network service providing centralized storage provisioning for iSCSI workloads. It supports Azure VMware Solution, AKS, and Azure VMs, offering **high throughput**, **sub-millisecond latency**, and **simplified management** compared to individual managed disks.

Q60.

How do you achieve zero RPO in Azure?

Answer:

Zero RPO is achievable using **synchronous replication**: Azure SQL Business Critical tier with Always On replicas, Cosmos DB strong consistency with multi-region writes, or **Zone-Redundant** services within a region. Cross-region zero RPO is replaced with near-zero RPO using asynchronous replication with very short intervals in practice.

Section 5: Infrastructure Design

Infrastructure design covers compute, containers, serverless, and hybrid architectures. AZ-305 tests your ability to select the right Azure compute services, design scalable and resilient infrastructure, and apply modern application architecture patterns.

Q61.

When would you choose Azure Kubernetes Service (AKS) over Azure App Service?

Answer:

Choose **AKS** for containerized microservices requiring fine-grained control over networking, scaling, runtime, and multi-container orchestration. Choose **Azure App Service** for web applications needing rapid deployment and managed platform services. AKS has higher operational complexity but greater control.

Q62.

What is Azure Virtual Machine Scale Sets (VMSS)?

Answer:

VMSS lets you create and manage a group of identical, load-balanced VMs that automatically scale in response to demand or a schedule. They support **uniform** and **flexible** orchestration modes and integrate with Azure Load Balancer, Application Gateway, and Azure Monitor for autoscaling.

Q63.

What is Azure App Service Plan?

Answer:

An App Service Plan defines the **region**, **number of VM instances**, and **VM size** for hosting App Service apps. Multiple apps can share a single plan. Tiers from Free/Shared to Isolated provide autoscaling, custom domains, SSL, and deployment slots at higher tiers.

Q64.

What is Azure Functions and what are its hosting plans?

Answer:

Azure Functions is a serverless compute service for event-driven code. Hosting plans: **Consumption** (pay per execution, auto-scales to zero), **Premium** (pre-warmed instances, VNet integration, no cold start), and **Dedicated (App Service Plan)** (predictable cost, always-on). Flex Consumption is the newest plan.

Q65.

What is an Azure App Service Environment (ASE)?

Answer:

ASE is a fully isolated and dedicated environment for securely running App Service apps at high scale inside a customer's VNet. It provides full network isolation, private IP addressing, and the ability to connect to on-premises resources, supporting the Isolated pricing tier.

Q66.

Explain blue-green deployment in Azure.

Answer:

Blue-green deployment uses two identical environments. **Blue** is the current production environment; **Green** is the new version. In Azure App Service, this uses **deployment slots**. After testing in the green slot, traffic is swapped with zero downtime. The old slot serves as an instant rollback option.

Q67.

What is Azure Container Apps?

Answer:

Azure Container Apps is a serverless container platform built on Kubernetes and KEDA. It abstracts infrastructure management while supporting **microservices**, **event-driven scaling**, **Dapr integration**, and **traffic splitting** for canary releases without managing the underlying cluster.

Q68.

What is Azure Dedicated Host?

Answer:

Azure Dedicated Host provides physical servers dedicated to a single customer's subscription. It gives visibility and control over the host's maintenance window, hardware isolation for compliance, and the ability to place VMs on a specific host for **BYOL licensing** and strict isolation requirements.

Q69.

What is the difference between Azure Logic Apps and Azure Functions?

Answer:

Azure Logic Apps is a low-code/no-code workflow automation service with 400+ built-in connectors, ideal for integrating enterprise applications. **Azure Functions** is a code-first serverless platform for event-driven compute. Logic Apps excels at orchestration; Functions excel at custom, complex processing logic.

Q70.

What is Azure Spot VMs?

Answer:

Azure Spot VMs provide access to unused Azure compute capacity at up to 90% discount. They can be evicted with 30 seconds notice when Azure needs capacity back. Suitable for **batch processing**, **dev/test**, **CI/CD pipelines**, and **fault-tolerant workloads**. Not suitable for production stateful workloads.

Q71.

Explain AKS node pools.

Answer:

AKS supports **system node pools** (run critical system pods like CoreDNS) and **user node pools** (run application workloads). Each pool can have different VM sizes, OS types, availability zones, and autoscaler configurations, enabling workload isolation and cost optimization using spot instances for non-critical workloads.

Q72.

What is the Durable Functions extension?

Answer:

Durable Functions enables writing stateful functions in a serverless environment. It implements patterns like **function chaining**, **fan-out/fan-in**, **async HTTP APIs**, and **human interaction** workflows. State is persisted in Azure Storage, enabling long-running orchestrations.

Q73.

How do you design for auto-scaling in Azure?

Answer:

Design for auto-scaling using: **VMSS** with scale rules based on CPU or custom metrics; **App Service autoscale** with schedule and metric-based rules; **AKS HPA** and Cluster Autoscaler; **Azure Functions Consumption plan** for serverless auto-scaling; and **Cosmos DB autoscale** for database throughput.

Q74.

What is Azure VMware Solution (AVS)?

Answer:

Azure VMware Solution allows organizations to run VMware workloads natively in Azure with dedicated, bare-metal hardware. It provides a fully managed VMware SDDC (vSphere, vSAN, NSX-T) integrated with Azure networking and services, supporting **lift-and-shift migrations** of existing VMware environments.

Q75.

What is Azure Spring Apps?

Answer:

Azure Spring Apps is a fully managed service for Spring Boot and Steeltoe (.NET) microservices. It handles service discovery (Eureka), configuration management (Config Server), distributed tracing, and CI/CD integration, allowing developers to focus on business logic while Azure manages the runtime infrastructure.

Section 6: Networking

Azure networking is a cornerstone of the AZ-305 exam. This section covers virtual networks, hybrid connectivity, load balancing, DNS, private endpoints, and network security. Architects must design secure, performant, and cost-effective network topologies for complex enterprise environments.

Q76.

What is an Azure Virtual Network (VNet)?

Answer:

A VNet is the fundamental building block of private networking in Azure. It provides isolation, segmentation, and communication for Azure resources. VNets contain **subnets**, support **custom DNS**, and can be connected via **peering**, **VPN Gateway**, or **ExpressRoute**, and to Azure services via **private endpoints**.

Q77.

What is the difference between VNet Peering and VPN Gateway?

Answer:

VNet Peering uses Azure's backbone network for low-latency, high-bandwidth VM-to-VM connectivity without encryption overhead. **VPN Gateway** uses IPsec/IKE encrypted tunnels supporting site-to-site, point-to-site, and VNet-to-VNet connectivity. Peering is preferred for Azure-to-Azure; VPN Gateway for on-premises connectivity.

Q78.

What is Azure ExpressRoute?

Answer:

ExpressRoute provides a private, dedicated connection from on-premises to Azure through a connectivity provider, bypassing the public internet. It offers **predictable latency**, **higher bandwidth** (50 Mbps to 100 Gbps), and **SLA-backed reliability**. ExpressRoute Global Reach enables connecting on-premises sites via Azure's backbone.

Q79.

What is an Azure Network Security Group (NSG)?

Answer:

An NSG is a virtual firewall containing security rules that control inbound and outbound network traffic. Rules are evaluated by **priority** (100-4096) and filtered by source/destination IP, port, and protocol. NSGs can be associated with **subnets** or individual **network interfaces**.

Q80.

What is Azure Private Endpoint?

Answer:

A Private Endpoint is a network interface with a private IP from your VNet connected to an Azure PaaS service (Storage, SQL, Key Vault). Traffic flows over the Azure backbone without traversing the public internet, providing **enhanced security** and enabling access from on-premises via VPN or ExpressRoute.

Q81.

What is the hub-and-spoke network topology?

Answer:

The **hub-and-spoke** topology uses a central hub VNet hosting shared services (firewall, VPN/ExpressRoute gateway, DNS) connected to multiple spoke VNets via peering. Spoke VNets host workloads and route traffic through the hub for inspection and security policy enforcement, centralizing security while isolating workloads.

Q82.

What is Azure Virtual WAN?

Answer:

Azure Virtual WAN is a networking service providing optimized and automated branch-to-branch and branch-to-Azure connectivity. It simplifies large-scale network management with a **managed hub** that aggregates VPN, ExpressRoute, and SD-WAN connections, ideal for enterprises with many branch offices needing global any-to-any connectivity.

Q83.

What is Azure Firewall?

Answer:

Azure Firewall is a managed, cloud-native network security service providing **stateful packet inspection**, **FQDN filtering**, **threat intelligence-based filtering**, and **DNAT/SNAT**. Azure Firewall Premium adds **TLS inspection**, **IDPS**, and **URL filtering** for advanced threat protection.

Q84.

What is Azure Application Gateway?

Answer:

Azure Application Gateway is a Layer 7 (HTTP/HTTPS) load balancer providing **URL-based routing**, **SSL/TLS termination**, **WebSocket support**, **cookie-based session affinity**, and built-in **WAF**. It is ideal for web applications requiring advanced routing and security at the application layer.

Q85.

What is Azure Load Balancer?

Answer:

Azure Load Balancer is a Layer 4 (TCP/UDP) load balancer that distributes inbound flows among healthy VM instances. **Public Load Balancer** handles inbound internet traffic. **Internal Load Balancer** handles internal traffic. It supports health probes, load balancing rules, outbound SNAT rules, and HA ports for NVAs.

Q86.

What is Azure DNS Private Zones?

Answer:

Azure DNS Private Zones provide DNS name resolution for resources within a VNet without public exposure. They can be linked to multiple VNets. **Auto-registration** automatically creates DNS records for VMs in linked VNets. Private zones are essential for private endpoint DNS resolution in enterprise designs.

Q87.

What is a User Defined Route (UDR)?

Answer:

A UDR is a custom route in Azure that overrides Azure's default system routes, added to a route table associated with a subnet. UDRs force traffic through a **network virtual appliance (NVA)**, **Azure Firewall**, or VPN Gateway, enabling centralized inspection and traffic flow control in hub-and-spoke topologies.

Q88.

What is Azure DDoS Protection?

Answer:

Azure DDoS Protection provides: **DDoS Network Protection** (Basic, automatic for all Azure services) and **DDoS Network Protection Standard** (adds adaptive tuning, attack telemetry, rapid response support, and SLA guarantee) protecting VNet resources against volumetric, protocol, and resource layer attacks.

Q89.

What is Azure NAT Gateway?

Answer:

Azure NAT Gateway provides outbound internet connectivity for resources in private subnets without individual public IPs. It uses **Source NAT (SNAT)** with public IP addresses or prefixes, offering predictable outbound connectivity and eliminating SNAT port exhaustion issues common with load balancer outbound rules.

Q90.

What is Azure Bastion?

Answer:

Azure Bastion is a fully managed PaaS providing secure RDP/SSH access to VMs directly from the Azure portal over TLS, without exposing VMs via public IPs or opening RDP/SSH ports. It eliminates the need for jump servers and integrates with Azure AD for passwordless access.

Section 7: Security and Monitoring

Security and monitoring are integrated throughout all AZ-305 exam domains. This section covers Microsoft Defender for Cloud, Azure Monitor, Key Vault, Sentinel, and security design patterns that architects must master to build secure, observable Azure solutions.

Q91.

What is Microsoft Defender for Cloud?

Answer:

Microsoft Defender for Cloud is a Cloud Security Posture Management (CSPM) and Cloud Workload Protection Platform (CWPP). It provides **Secure Score**, **security recommendations**, **threat detection**, **regulatory compliance dashboards**, and protection for servers, databases, containers, storage, and more.

Q92.

What is Azure Monitor?

Answer:

Azure Monitor is the comprehensive monitoring platform for Azure and hybrid environments. It collects **metrics** (numerical, time-series data) and **logs** (structured records) from resources, enabling analysis via **Metrics Explorer** and **Log Analytics**. It supports alerts, dashboards, workbooks, and integrates with Application Insights.

Q93.

What is Azure Key Vault?

Answer:

Azure Key Vault is a managed service for securely storing and accessing **secrets** (connection strings, passwords), **keys** (cryptographic keys), and **certificates** (SSL/TLS). It provides access logging, RBAC and access policies, HSM-backed storage (Premium tier), and soft delete with purge protection for compliance.

Q94.

What is Microsoft Sentinel?

Answer:

Microsoft Sentinel is a cloud-native SIEM and SOAR solution. It ingests data from diverse sources, uses ML-based analytics to detect threats, and enables automated response via **playbooks** (Logic Apps). It scales elastically without infrastructure management, providing unified security operations.

Q95.

What is Azure Application Insights?

Answer:

Application Insights is an APM service within Azure Monitor providing **distributed tracing**, **dependency tracking**, **exception logging**, **user behavior analytics**, **availability tests**, and **live metrics**. Instrumentation is via SDK, OpenTelemetry, or auto-instrumentation for supported platforms.

Q96.

What is a Log Analytics Workspace?

Answer:

A Log Analytics Workspace is the central repository for log data in Azure Monitor. Resources send diagnostic logs and metrics to the workspace, queried using **KQL (Kusto Query Language)**. Defender for Cloud, Sentinel, and VM Insights all use Log Analytics as their underlying data store.

Q97.

What is the Zero Trust security model?

Answer:

Zero Trust is a security framework based on '**Never trust, always verify**.' It assumes breach and requires explicit verification of every request regardless of network location. Pillars include **identity**, **endpoints**, **applications**, **data**, **infrastructure**, and **network** — each independently verified before granting access.

Q98.

What is Microsoft Defender for Containers?

Answer:

Microsoft Defender for Containers provides security for Kubernetes environments including AKS and Arc-enabled clusters. It provides **vulnerability assessments** for container images, **real-time threat detection**, **Kubernetes audit log analysis**, and **security recommendations** aligned to CIS benchmarks.

Q99.

What is Azure Policy Deny effect?

Answer:

The **Deny** effect in Azure Policy prevents a resource deployment or modification that doesn't comply with the policy rule — for example, blocking storage accounts without HTTPS-only enabled. Deny is evaluated before resource creation, unlike Audit which only flags non-compliance after deployment.

Q100.

How do you design secrets management in Azure?

Answer:

Best practices: store all secrets in **Azure Key Vault**, use **Managed Identities** for access without credentials in code, enable **soft delete and purge protection**, restrict access using **RBAC**, enable **diagnostic logging** to Log Analytics, and rotate secrets regularly using Key Vault's automatic rotation feature.

Q101.

What is Microsoft Defender for SQL?

Answer:

Microsoft Defender for SQL provides advanced threat protection for Azure SQL Database, SQL Managed Instance, and SQL Server on VMs. It detects anomalous activities indicating harmful attempts including **SQL injection**, **brute force**, and **unusual access patterns**, sending alerts with investigation guidance.

Q102.

What is Azure Monitor Alerts?

Answer:

Azure Monitor Alerts proactively notify you when conditions indicate potential issues. Types include **Metric Alerts** (threshold-based), **Log Alerts** (KQL query-based), **Activity Log Alerts** (resource operations), and **Smart Detection**. Alerts trigger **Action Groups** to notify or automate responses.

Q103.

What is the shared responsibility model in Azure?

Answer:

Microsoft manages **physical datacenter security, hardware, and the hypervisor**. Customers are always responsible for **data, identities, and access management**. For IaaS, customers additionally manage OS, runtime, and applications. For PaaS/SaaS, Microsoft takes more responsibility up the stack.

Q104.

What is Microsoft Defender for Storage?

Answer:

Microsoft Defender for Storage detects unusual and potentially harmful attempts to access or exploit Azure storage accounts, including **sensitive data exposure**, **malware uploads**, **anonymous access attempts**, and **suspicious access patterns**. It integrates with Sentinel and can trigger Logic Apps for automated response.

Q105.

What is Azure Monitor Workbooks?

Answer:

Azure Monitor Workbooks are interactive reports combining text, KQL queries, metrics, parameters, and rich visualizations. They support **custom templates**, **parameterized queries**, and **cross-resource queries**. Widely used for security dashboards, compliance reporting, and operational runbooks.

Extended Q&A; Bank — Azure Architecture Design (Part 2)

Q1.

What is the CAP theorem and how does it apply to Azure services?

Answer:

The CAP theorem states that a distributed system can provide at most two of: **Consistency**, **Availability**, and **Partition tolerance**. Azure Cosmos DB lets you choose your balance via consistency levels. Azure SQL Database prioritizes CP (consistency + partition tolerance), while storage queues prioritize AP.

Q2.

What is Azure Cost Management?

Answer:

Azure Cost Management + Billing provides tools to monitor, allocate, and optimize Azure spending. Features include **cost analysis**, **budgets and alerts**, **cost recommendations** from Advisor, **export to storage**, and **showback/chargeback** for internal billing using tags and management group scopes.

Q3.

What are Azure Management Groups?

Answer:

Management Groups are containers for organizing subscriptions above the subscription level. They enable applying **Azure Policy** and **RBAC** at scale across multiple subscriptions. Policies applied at a parent management group are inherited by all child groups and subscriptions below it.

Q4.

Explain the concept of idempotency in cloud architectures.

Answer:

Idempotency means performing the same operation multiple times produces the same result as performing it once. Critical for **retry logic**, **message processing**, and **distributed transactions**. Azure Storage and Cosmos DB support conditional writes (ETags) to implement idempotent operations safely.

Q5.

What is the strangler fig pattern in cloud migration?

Answer:

The strangler fig pattern gradually replaces a legacy system by routing new functionality to a cloud-based system while the legacy handles remaining requests. Over time, the new system 'strangles' the old until it's decommissioned. **Azure API Management** is commonly used as the facade for routing traffic during migration.

Q6.

What is Azure Lighthouse?

Answer:

Azure Lighthouse enables managed service providers (MSPs) to manage customer Azure environments at scale from their own tenant using **Azure delegated resource management**, providing cross-tenant visibility and control without requiring elevated permissions in each customer tenant.

Q7.

What is the CQRS pattern?

Answer:

CQRS (Command Query Responsibility Segregation) separates read and write operations into different models. The **command model** handles writes; the **query model** handles reads from a denormalized read store. In Azure, Event Grid or Service Bus for commands and Cosmos DB or Redis for the read model implement this pattern.

Q8.

How do you implement a tagging strategy in Azure?

Answer:

Define **mandatory tags** (environment, owner, cost-center, project) enforced via Azure Policy Deny effect, use **initiative definitions** to group tag policies, automate tag inheritance using **Modify effect policies**, and use tags for cost analysis, automation triggers, and RBAC scoping.

Q9.

What is Azure Arc?

Answer:

Azure Arc extends Azure management capabilities to resources outside Azure, including on-premises servers, Kubernetes clusters, SQL servers, and VMware/Hyper-V VMs. Arc-enabled resources appear in the Azure portal and support **Azure Policy**, **Microsoft Defender**, **Azure Monitor**, and **GitOps**.

Q10.

What is an Azure Private DNS Resolver?

Answer:

Azure DNS Private Resolver is a managed service enabling on-premises clients to resolve Azure Private DNS zone names and vice versa. It provides **inbound endpoints** (for on-premises to Azure DNS queries) and **outbound endpoints** (for Azure to on-premises DNS queries), replacing custom DNS VMs in hybrid designs.

Q11.

What is the bulkhead pattern?

Answer:

The bulkhead pattern isolates elements of an application into pools so that if one fails, the others continue to function. In Azure microservices, this means separate **App Service Plans**, **AKS node pools**, or **Service Bus queues** per service, preventing a single failing component from consuming shared resources.

Q12.

What is the circuit breaker pattern?

Answer:

The circuit breaker prevents an application from repeatedly attempting an operation likely to fail. After a threshold of failures, the circuit 'opens' and calls fail immediately. After a timeout, it enters 'half-open' to test recovery. **Azure API Management policies** or the **Polly** library implement circuit breaking.

Q13.

What is Bicep?

Answer:

Bicep is a domain-specific language that transpiles to ARM JSON templates, providing cleaner syntax for Azure IaC. It supports **modules**, **loops**, **conditions**, **type safety**, and **IDE tooling**. It is Microsoft's recommended approach for ARM-native IaC, offering better readability than raw ARM JSON.

Q14.

What is Azure Traffic Manager?

Answer:

Azure Traffic Manager is a DNS-based global load balancer routing client requests to endpoints based on routing methods: **Priority**, **Weighted**, **Performance**, **Geographic**, **Multivalue**, and **Subnet**. It monitors endpoint health and automatically redirects traffic on failure.

Q15.

What is Azure Chaos Studio?

Answer:

Azure Chaos Studio is a managed service for improving application resilience by intentionally introducing faults (chaos experiments). It supports **service-direct faults** (VM shutdown, network disconnect) and **agent-based faults** (CPU pressure, memory pressure), helping validate DR plans and failover mechanisms.

Q16.

What is Azure API Management (APIM)?

Answer:

APIM is a managed API gateway providing **authentication** (OAuth, JWT validation), **rate limiting**, **caching**, **request/response transformation**, **logging**, and a **developer portal**. It supports backends in Azure, on-premises, or any cloud via VNet integration.

Q17.

What is Azure Event Grid?

Answer:

Azure Event Grid is a fully managed event routing service using a pub/sub model. It routes events from **Azure services** or custom sources to **handlers** (Azure Functions, Logic Apps, Event Hubs, webhooks), providing at-least-once delivery, retry policies, and dead-letter support.

Q18.

What is the sidecar pattern?

Answer:

The sidecar pattern deploys a helper container alongside the main application container to handle cross-cutting concerns like **logging**, **monitoring**, **configuration**, **service mesh proxy**, or **secrets injection** without modifying the main application. Common in AKS with Dapr or Istio sidecars.

Q19.

What is Azure Container Registry (ACR)?

Answer:

ACR is a managed, private container registry service. Features include **geo-replication**, **ACR Tasks** for automated builds, **content trust** for image signing, **vulnerability scanning** via Defender for Containers, and **private endpoint** support for network-isolated container image storage.

Q20.

What is the retry pattern with exponential backoff?

Answer:

The retry pattern re-executes a failed operation automatically. **Exponential backoff** increases the delay between retries exponentially (e.g., 1s, 2s, 4s, 8s) plus random jitter to prevent thundering herd effects. Azure SDKs have this built-in; the **Polly** library provides retry policies for custom .NET code.

Q21.

What is Azure Deployment Environments?

Answer:

Azure Deployment Environments enables dev teams to self-serve sandbox environments from infrastructure templates (ARM, Bicep, Terraform). Platform engineers define **environment types** with governance controls, while developers provision full application environments on-demand, improving velocity while maintaining compliance.

Q22.

What is KEDA?

Answer:

KEDA (Kubernetes Event-Driven Autoscaling) scales containerized workloads based on external event sources beyond CPU/memory. In Azure, KEDA scales on Service Bus queue depth, Event Hub lag, Blob Storage blob count, or HTTP traffic. Azure Container Apps uses KEDA natively for event-driven autoscaling.

Q23.

What is Azure Managed Applications?

Answer:

Azure Managed Applications enable vendors to publish cloud solutions with a locked-down resource group that customers cannot modify. The publisher maintains management access. Used by ISVs to offer managed SaaS-like experiences on Azure while protecting proprietary deployment configurations.

Q24.

What is the outbox pattern?

Answer:

The outbox pattern ensures reliable message publishing alongside database transactions. Instead of publishing events directly, the service writes events to an **outbox table** in the same transaction as business data. A separate process reads the outbox and publishes to the message broker, ensuring consistency without distributed transactions.

Q25.

What is Azure Well-Architected Review?

Answer:

The Azure Well-Architected Review (available at aka.ms/assessments) evaluates workloads against the five WAF pillars. It asks targeted questions and produces a prioritized list of recommendations, helping teams identify gaps in their current Azure architecture aligned to Microsoft best practices.

Q26.

What is Azure Service Fabric?

Answer:

Azure Service Fabric is a distributed systems platform for deploying and managing scalable, reliable microservices. It provides **stateful services** with built-in state replication, **actor model**, **rolling upgrades**, and health monitoring. It is the platform underlying many internal Microsoft services.

Q27.

What is Azure Spring Apps Enterprise tier?

Answer:

Azure Spring Apps Enterprise tier provides VMware Spring Runtime support including **Tanzu Build Service** (automatic builds without Dockerfiles), **Spring Cloud Gateway** (managed API gateway), **API Portal**, and **Application Configuration Service**, targeting production Spring microservices with enterprise SLAs.

Q28.

What is the ambassador pattern?

Answer:

The ambassador pattern deploys a proxy service alongside an application to offload cross-cutting tasks like **retries**, **circuit breaking**, **authentication**, and **monitoring**. In Azure, this can be implemented as a sidecar in AKS or as a separate Azure API Management layer in front of backend services.

Q29.

What is Azure Policy Guest Configuration?

Answer:

Azure Policy Guest Configuration (now Azure Automanage Machine Configuration) audits and enforces settings inside VMs, both Azure and Arc-enabled. It checks OS configurations, software presence, registry settings, and security baselines using PowerShell DSC or Chef InSpec definitions without requiring custom agents.

Q30.

What is Azure Web PubSub?

Answer:

Azure Web PubSub is a fully managed real-time messaging service enabling bidirectional communication between server and clients using WebSocket or other web protocols. It supports **pub/sub messaging**, **group broadcasting**, and **connection management** at scale for real-time dashboards, chat, and live collaboration.

Extended Q&A; Bank — Identity & Governance (Part 2)

Q1.

What is Cross-Tenant Access Settings in Azure AD?

Answer:

Cross-tenant access settings control how your Azure AD tenant collaborates with external tenants. **Inbound settings** control external user access to your resources. **Outbound settings** control your users' access to external tenants, enabling granular trust relationships for MFA and device compliance claims.

Q2.

What is Azure AD Domain Services (AADDS)?

Answer:

Azure AD Domain Services provides managed domain services like LDAP, Kerberos, NTLM, and Group Policy in Azure without deploying domain controllers. It synchronizes identities from Azure AD and supports domain-joining Azure VMs for legacy applications requiring traditional AD features without on-premises infrastructure.

Q3.

What is Microsoft Entra Permissions Management?

Answer:

Microsoft Entra Permissions Management is a CIEM tool that discovers, remediates, and monitors identity permissions across Azure, AWS, and GCP. It detects **over-provisioned identities**, identifies **unused permissions**, and provides a **Permission Creep Index (PCI)** score to assess risk across multi-cloud environments.

Q4.

What is Continuous Access Evaluation (CAE)?

Answer:

CAE enables Azure AD to revoke access tokens in near real-time when critical events occur (user disabled, password changed, IP change). CAE-capable applications receive events and immediately re-authenticate, reducing the window for compromised token abuse compared to waiting for standard token expiry.

Q5.

What is Azure AD Named Locations?

Answer:

Named Locations define IP ranges or countries considered trusted in Conditional Access policies. They can **block access from specific countries**, **require MFA from non-trusted IPs**, or **allow access only from office IP ranges**, serving as key building blocks for location-based Conditional Access policies.

Q6.

What is the principle of separation of duties in Azure?

Answer:

Separation of duties prevents a single person from having enough privilege to abuse a system. In Azure: separate **subscription owners** from **security administrators**, use **PIM** to require approval for sensitive roles, implement **dual authorization** for critical operations, and use **access reviews** to validate ongoing access.

Q7.

What is Microsoft Entra Workload Identities?

Answer:

Microsoft Entra Workload Identities manages identities for applications, services, and automated processes. It includes **service principal lifecycle management**, **credential hygiene** (detecting unused or expiring secrets), **Conditional Access for workload identities**, and **identity risk detection** for non-human identities.

Q8.

What is SCIM provisioning in Azure AD?

Answer:

SCIM (System for Cross-domain Identity Management) automates user and group provisioning between Azure AD and SaaS applications. Azure AD's **automatic user provisioning** supports SCIM 2.0, creating, updating, and deactivating accounts in connected apps based on Azure AD lifecycle events, eliminating manual provisioning.

Q9.

What is the difference between authentication and authorization?

Answer:

Authentication verifies the identity of a user or service (who are you?) using credentials, certificates, or tokens. **Authorization** determines what an authenticated principal is allowed to do (what can you do?). In Azure, **Azure AD** handles authentication and **Azure RBAC** handles authorization.

Q10.

What is an Azure AD access review?

Answer:

Access reviews in Azure AD enable periodic review of user access to groups, applications, and privileged roles. Reviewers confirm or revoke access; automated actions can remove access when no response is received. Access reviews support compliance requirements for **SOX**, **ISO 27001**, and similar frameworks.

Q11.

What is passwordless authentication in Azure AD?

Answer:

Azure AD supports three passwordless methods: **Windows Hello for Business** (biometric/PIN tied to device), **FIDO2 security keys** (hardware keys like YubiKey), and **Microsoft Authenticator app** (phone sign-in). All replace passwords with cryptographic credentials, dramatically reducing phishing risk.

Q12.

What is an Administrative Unit in Azure AD?

Answer:

Administrative Units restrict administrative scope to a subset of users, groups, or devices. They enable role delegation at a sub-tenant level, such as allowing a **Helpdesk Administrator** for only one department's users. Useful in large organizations where IT management is decentralized by department or region.

Q13.

What is Managed HSM in Azure Key Vault?

Answer:

Azure Key Vault Managed HSM is a fully managed, single-tenant FIPS 140-2 Level 3 validated Hardware Security Module service. It provides **exclusive control** over cryptographic keys without Microsoft access, supports **local RBAC**, and meets stringent compliance requirements for financial services and healthcare.

Q14.

What is federated identity credentials?

Answer:

Federated identity credentials enable workloads in external environments (GitHub Actions, Kubernetes, AWS) to authenticate to Azure without managing secrets. The external IdP issues a token that Azure AD accepts based on a **trust relationship**, exchanging it for an Azure AD token, eliminating long-lived secrets in CI/CD pipelines.

Q15.

What is Defender CSPM?

Answer:

Defender CSPM provides **attack path analysis**, **cloud security explorer**, **agentless scanning**, **data sensitivity context**, and **governance** capabilities. It identifies the highest-risk attack paths across multi-cloud environments and prioritizes remediation efforts based on actual exploitability.

Q16.

What is Just-In-Time VM Access?

Answer:

JIT VM Access in Microsoft Defender for Cloud reduces attack surface by locking down inbound management ports (RDP/SSH). When access is needed, users request it through Defender for Cloud, which temporarily opens the required port for a limited time and specific source IP. All access requests are logged for audit.

Q17.

What are Azure resource locks?

Answer:

Resource locks prevent accidental modification or deletion of Azure resources. **CanNotDelete** allows modifications but prevents deletion. **ReadOnly** prevents both modification and deletion. Locks are inherited by child resources and override RBAC — even subscription owners cannot delete a locked resource without first removing the lock.

Q18.

What is Azure Confidential Computing?

Answer:

Azure Confidential Computing protects data while it is **in use** (being processed) using hardware-based Trusted Execution Environments (TEEs) like Intel SGX and AMD SEV-SNP. Used in healthcare, financial services, and government workloads where data must remain protected even from cloud operators.

Q19.

What is Microsoft Purview Information Protection?

Answer:

Microsoft Purview Information Protection provides data classification, labeling, and protection across Microsoft 365 and Azure. **Sensitivity labels** apply protections (encryption, watermarks, access restrictions). Labels can be applied manually or auto-applied based on content inspection, extending protection across cloud and on-premises.

Q20.

What is the difference between a service principal and a managed identity?

Answer:

A **service principal** is manually created, requires managing credentials (secrets or certificates), and can be used from any platform. A **managed identity** is automatically managed by Azure — no credential management required — but can only be used from Azure resources. Managed identities are strongly preferred for Azure-to-Azure authentication.

Q21.

What is the Microsoft cloud security benchmark?

Answer:

The Microsoft Cloud Security Benchmark (MCSB) provides security recommendations based on CIS and NIST frameworks, tailored for Azure and multi-cloud. It covers **network security**, **identity management**, **data protection**, and **asset management**. Defender for Cloud maps its recommendations to MCSB for compliance tracking.

Q22.

What is Azure Subscription Vending?

Answer:

Subscription Vending is an automated process for provisioning Azure subscriptions with standardized governance controls. Using Azure Landing Zones and IaC, new subscriptions are automatically deployed with required **management group placement**, **policy assignments**, **RBAC**, **networking**, and **monitoring** configurations.

Q23.

What is the difference between security defaults and Conditional Access?

Answer:

Security defaults provide a simple, one-click set of basic security controls (MFA for all users, block legacy auth) suitable for smaller organizations. **Conditional Access** provides granular, risk-based policies requiring Azure AD P1/P2 licenses, enabling fine-tuned control over access based on user, device, location, and risk.

Q24.

What is Azure AD Token Lifetime Policy?

Answer:

Token Lifetime Policy configures the lifetime of access tokens, ID tokens, and SAML tokens in Azure AD. For security, combine short token lifetimes with **Continuous Access Evaluation (CAE)**, which provides real-time revocation without sacrificing usability, rather than relying solely on short expiry windows.

Q25.

What is Microsoft Defender for Identity architecture?

Answer:

Microsoft Defender for Identity uses a **sensor** installed on domain controllers capturing and analyzing AD traffic locally, sending metadata to the cloud service for ML analysis. It monitors **LDAP queries**, **Kerberos tickets**, **NTLM authentication**, and **AD replication** for attack indicators across the customer base.

Q26.

What is Azure Policy Remediation?

Answer:

Azure Policy Remediation brings non-compliant resources into compliance after a policy is assigned. For policies with **Modify** or **DeployIfNotExists** effects, a remediation task creates a managed identity that applies required changes to existing resources. Remediation can be manually triggered or configured for automatic remediation.

Q27.

What is a Privileged Access Workstation (PAW)?

Answer:

A PAW is a dedicated, hardened workstation used exclusively for administrative tasks. In Azure, PAW implementation includes **dedicated admin accounts**, **PIM for just-in-time access**, **Conditional Access requiring compliant devices**, and **Microsoft Defender for Endpoint** enforcement on the admin device.

Q28.

What is Azure Confidential Ledger?

Answer:

Azure Confidential Ledger is a tamper-proof, append-only ledger service built on blockchain running in Azure Confidential Computing SGX enclaves. It provides cryptographically verifiable records with no Microsoft access, suitable for **audit trails**, **supply chain transparency**, and **financial records** requiring immutability.

Q29.

What is Microsoft Entra Verified ID?

Answer:

Microsoft Entra Verified ID enables issuance and verification of digital credentials based on open standards (W3C Verifiable Credentials, DID). Organizations issue verifiable credentials (e.g., employee ID, degree) that users store in a digital wallet and present for verification without the issuer being involved in the verification.

Q30.

What is Azure AD Sign-in Logs?

Answer:

Sign-in Logs record all authentication events in Azure AD including interactive sign-ins, non-interactive sign-ins, managed identity sign-ins, and service principal sign-ins. They capture success/failure, location, MFA result, Conditional Access outcomes, and risk level. Logs are retained for 30 days and exportable to Log Analytics.

Extended Q&A; Bank – Data Storage (Part 2)

Q1.

What is the difference between LRS and GRS?

Answer:

LRS (Locally Redundant Storage) replicates data three times within a single datacenter (99.999999999% durability). **GRS (Geo-Redundant Storage)** adds asynchronous replication to a paired region (16 nines durability). LRS is less expensive; GRS protects against regional disasters. RA-GRS additionally allows reading from the secondary.

Q2.

What is Azure Managed Disks?

Answer:

Azure Managed Disks are block storage volumes for Azure VMs managed by Azure. Types: **Ultra Disk** (highest IOPS, mission-critical), **Premium SSD v2** (flexible performance), **Premium SSD**, **Standard SSD**, and **Standard HDD**. Managed Disks support snapshots, encryption, and availability zone pinning.

Q3.

What is Azure Blob Storage lifecycle management?

Answer:

Lifecycle management policies automate transitioning blobs between access tiers and deleting old blobs. Rules evaluate **last modified time**, **last accessed time**, or **creation time**. Example: move to Cool after 30 days, Archive after 90 days, delete after 365 days — reducing storage costs automatically.

Q4.

What is the difference between Azure SQL Database and SQL Managed Instance?

Answer:

Azure SQL Database is best for new cloud applications with hyperscale support. **SQL Managed Instance** offers near 100% SQL Server compatibility including SQL Agent, linked servers, and CLR — ideal for migrating existing SQL Server applications with minimal code changes.

Q5.

What is Cosmos DB change feed?

Answer:

The Cosmos DB change feed is an ordered log of all inserts and updates to a container. It enables **event-driven processing**, **real-time data movement**, **cache invalidation**, and **event sourcing**. Consumed via Azure Functions (Cosmos DB trigger) or the Change Feed Processor SDK.

Q6.

What is Azure Synapse Link?

Answer:

Azure Synapse Link integrates operational databases (Cosmos DB, Azure SQL Database) with Azure Synapse Analytics, enabling **hybrid transactional/analytical processing (HTAP)**. Run analytical queries on live operational data without ETL pipelines or impacting production, using auto-synced analytical stores.

Q7.

What is immutable blob storage?

Answer:

Immutable blob storage allows writing blobs to a WORM (Write Once, Read Many) state. **Time-based retention policies** prevent deletion for a set duration. **Legal hold policies** retain data indefinitely. Immutable storage meets SEC 17a-4, CFTC, FINRA, and other financial regulatory requirements.

Q8.

What is the difference between Azure Event Hubs and Azure Service Bus?

Answer:

Azure Event Hubs is a high-throughput event streaming platform (millions of events/second) optimized for telemetry and real-time analytics with event replay. **Azure Service Bus** is an enterprise message broker for transactional messaging supporting ordered delivery, sessions, dead-letter queues, and request-reply patterns.

Q9.

What is Microsoft Purview for data governance?

Answer:

Microsoft Purview provides a unified data governance platform for discovering, classifying, and managing data across on-premises, Azure, and multi-cloud sources. It builds an **enterprise data map**, provides **data lineage tracking**, enables **business glossary** management, and supports self-service data discovery.

Q10.

What is Azure Stream Analytics?

Answer:

Azure Stream Analytics is a real-time analytics service for processing high-velocity data streams using a **SQL-like query language** with time-windowing functions (tumbling, hopping, sliding). Inputs include Event Hubs, IoT Hub, and Blob Storage; outputs include Power BI, SQL Database, and Functions.

Q11.

What is Azure Data Share?

Answer:

Azure Data Share enables organizations to securely share data with external partners without complex pipelines. It supports **snapshot-based sharing** (periodic snapshots) and **in-place sharing** (real-time access). Supported sources include ADLS Gen2, Azure SQL Database, Synapse, and Blob Storage.

Q12.

What is Azure Database Migration Service?

Answer:

Azure Database Migration Service (DMS) is a fully managed migration service supporting migrations from SQL Server, MySQL, PostgreSQL, MongoDB, and Oracle to Azure targets with both **online (minimal downtime)** and **offline** migration modes.

Q13.

What is Azure Disk Encryption?

Answer:

Azure Disk Encryption encrypts OS and data disks of Azure VMs using **BitLocker** (Windows) or **dm-crypt** (Linux) with keys stored in Azure Key Vault. It differs from Storage Service Encryption (SSE) which encrypts at the storage layer — ADE encrypts within the VM, protecting data from storage infrastructure access.

Q14.

What is the difference between point-in-time restore and geo-restore in Azure SQL?

Answer:

Point-in-time restore (PITR) restores an Azure SQL Database to any point within the backup retention period (1-35 days). **Geo-restore** restores from a geo-redundant backup to any Azure region when the primary region is unavailable, with a higher RPO of up to 1 hour compared to PITR.

Q15.

What is Azure NetApp Files?

Answer:

Azure NetApp Files is an enterprise-grade, high-performance file storage service powered by NetApp technology. It provides NFS and SMB protocol support with **ultra-low latency**, **high IOPS**, and **dynamic service levels** (Standard, Premium, Ultra). Used for SAP HANA, HPC, VDI, and latency-sensitive workloads.

Q16.

What is Cosmos DB serverless?

Answer:

Cosmos DB serverless is a consumption-based pricing mode where you pay only for consumed RUs, with no provisioned throughput. Ideal for **dev/test**, **low-traffic applications**, and **bursty workloads**. It does not support multi-region writes or the analytical store, making it less suitable for globally distributed production workloads.

Q17.

What is storage account network rules?

Answer:

Storage account network rules restrict access to specific **virtual networks** (via service endpoints or private endpoints), **IP address ranges**, and **trusted Azure services**. Default action can be set to Deny, ensuring only explicitly allowed sources can access the storage account.

Q18.

What is the difference between Azure Databricks and Azure Synapse Spark?

Answer:

Both are Apache Spark services. **Azure Databricks** offers superior ML/AI capabilities, MLflow, Delta Lake, and collaborative notebooks. **Azure Synapse Spark** integrates tightly with the Synapse workspace with shared metadata and native Synapse SQL connectivity. Databricks is preferred for ML; Synapse for unified analytics.

Q19.

What is Azure Elastic Pool?

Answer:

Azure SQL Elastic Pool enables multiple databases to share a pool of resources (DTUs or vCores), providing cost-effective solutions for databases with unpredictable usage patterns. Databases can burst to use more resources when needed while shared capacity keeps costs lower than individual provisioning.

Q20.

What is Azure Confidential Ledger for storage?

Answer:

Azure Confidential Ledger provides tamper-proof, append-only storage for audit records and sensitive data using blockchain technology in SGX enclaves. It provides cryptographically verifiable records suitable for **audit trails**, **supply chain**, and **financial records** requiring immutability and verifiable integrity.

Extended Q&A; Bank — Infrastructure Design (Part 2)

Q1.

What is the difference between Azure Container Instances and Azure Container Apps?

Answer:

Azure Container Instances (ACI) runs individual containers on-demand without orchestration — ideal for batch jobs and CI/CD tasks. **Azure Container Apps** provides a full microservices platform with autoscaling, ingress, Dapr integration, and traffic splitting, ideal for complex multi-container applications.

Q2.

What is Azure Container Apps Jobs?

Answer:

Container Apps Jobs run containerized tasks that execute and complete rather than run continuously. They support **manual** (on-demand), **scheduled** (cron-based), and **event-driven** (KEDA-triggered) execution modes. Ideal for batch processing, ETL jobs, data migration, and CI/CD pipelines.

Q3.

What is Azure Kubernetes Fleet Manager?

Answer:

Azure Kubernetes Fleet Manager enables managing multiple AKS clusters as a fleet. It provides **multi-cluster workload scheduling**, **fleet-wide Kubernetes version upgrades**, **cross-cluster load balancing**, and a unified control plane for large-scale AKS deployments across regions and subscriptions.

Q4.

What is Azure Red Hat OpenShift (ARO)?

Answer:

ARO is a fully managed OpenShift service jointly operated by Microsoft and Red Hat. It provides a fully managed Kubernetes platform with **OpenShift developer tooling**, **integrated CI/CD**, **operator framework**, and **Red Hat support** for organizations standardized on OpenShift.

Q5.

What is the difference between Azure Functions Flex Consumption and Premium plan?

Answer:

Flex Consumption supports per-instance concurrency, faster scaling, VNet integration, and per-execution billing — combining scale-to-zero with Premium features. **Premium plan** uses pre-warmed instances (no cold starts), supports VNet integration, and bills per instance even when idle.

Q6.

What is AKS Workload Identity?

Answer:

AKS Workload Identity uses **Federated Identity Credentials** and Kubernetes service account token projection to allow pods to authenticate to Azure services using Managed Identities without secrets. It follows the same federated identity standard as GitHub Actions, replacing pod identity (AAD Pod Identity).

Q7.

What is Azure App Configuration?

Answer:

Azure App Configuration centralizes application settings and feature flags separately from code. It provides **point-in-time snapshot** of configurations, **feature flag management** with percentage rollout, **dynamic configuration refresh** without redeployment, and Key Vault references for secrets. Ideal for microservices architectures.

Q8.

What is Azure Image Builder?

Answer:

Azure Image Builder is a managed service for creating custom VM images using a JSON template. It supports builds from Azure Marketplace base images, existing custom images, or RHSM content. Images can include pre-installed software, configurations, and scripts, distributed to Shared Image Gallery for multi-region deployment.

Q9.

What is Azure Shared Image Gallery?

Answer:

Azure Compute Gallery stores and manages custom VM images and application packages across regions and subscriptions. It supports **image versioning**, **global replication**, **RBAC sharing**, and **community sharing**, reducing deployment time by keeping images close to compute regions.

Q10.

What is Azure DevTest Labs?

Answer:

Azure DevTest Labs enables developers and testers to quickly create environments with minimal overhead. Lab owners define **allowed VM sizes**, **maximum VMs per user**, **auto-shutdown schedules**, and **artifact libraries**. It reduces waste by automatically shutting down idle environments and controlling costs.

Q11.

What is Azure Automanage?

Answer:

Azure Automanage automatically applies Azure best practices for VMs through their entire lifecycle, enabling and configuring services including **Azure Backup**, **Azure Monitor**, **Microsoft Defender for Cloud**, **Guest Configuration**, and **Update Management** based on Development or Production profiles.

Q12.

What is the difference between Azure Logic Apps Standard and Consumption?

Answer:

Consumption Logic Apps are multi-tenant, serverless, billed per execution, with a single workflow per app. **Standard** Logic Apps run on single-tenant infrastructure, support **multiple workflows per app**, **VNet integration**, **private endpoints**, and **stateful/stateless modes**, suitable for enterprise requirements.

Q13.

What is AKS node auto-provisioning?

Answer:

AKS node auto-provisioning (powered by Karpenter) automatically selects the optimal VM SKU for pending pods based on resource requests, labels, and node affinity rules. Unlike Cluster Autoscaler, it dynamically creates node pools with the right VM types, improving bin-packing efficiency and reducing cost.

Q14.

What is Azure Load Testing?

Answer:

Azure Load Testing is a fully managed service that generates high-scale load for validating application performance. It uses **Apache JMeter** scripts, runs tests from multiple regions, provides real-time metrics, integrates with Azure Monitor and Application Insights, and supports CI/CD pipeline integration for automated performance validation.

Q15.

What is Azure Static Web Apps?

Answer:

Azure Static Web Apps automatically builds and deploys full-stack web applications from GitHub or Azure DevOps. It provides **globally distributed hosting** for static assets, **managed API backends** via Azure Functions, **built-in authentication** (Azure AD, GitHub, Twitter), **custom domains**, and free SSL certificates.

Q16.

What is the difference between scale sets Uniform and Flexible orchestration?

Answer:

Uniform orchestration uses identical VMs from a single VM profile, optimized for stateless, large-scale workloads. **Flexible orchestration** allows mixing VM types, using individual VM management APIs, and supports availability zones. Flexible is recommended for new deployments for greater flexibility.

Q17.

What is Azure Service Bus dead-letter queue?

Answer:

The dead-letter queue (DLQ) is a sub-queue where Service Bus moves messages that cannot be delivered after the maximum delivery count is exceeded. Messages in the DLQ retain their properties for analysis and reprocessing. Monitoring DLQ depth is essential for detecting processing failures.

Q18.

What is the difference between Azure Cache for Redis Basic, Standard, and Premium?

Answer:

Basic: Single node, no SLA, dev/test. **Standard:** Two nodes (primary + replica), 99.9% SLA, automatic failover. **Premium:** Adds clustering, VNet injection, geo-replication, persistence, and zone redundancy. Enterprise tiers add Redis Modules and 99.999% SLA.

Q19.

What is Azure CycleCloud?

Answer:

Azure CycleCloud orchestrates HPC clusters in Azure, supporting schedulers like SLURM, PBS, Grid Engine, and LSF. It enables auto-scaling HPC clusters based on job queue demand using Spot VMs for cost efficiency, used by research institutions and enterprises running simulations and scientific computing.

Q20.

What is Azure SignalR Service?

Answer:

Azure SignalR Service is a fully managed real-time communication service that offloads WebSocket connections from application servers. It enables **server push** to web clients for real-time features like live dashboards, notifications, and multiplayer games, supporting 100,000+ concurrent connections per instance.

Q21.

What is Azure Orbital?

Answer:

Azure Orbital is a ground station as a service that enables communicating with satellites directly from Azure. It provides satellite scheduling, downlink/uplink contact, and data processing using Azure services. Used in earth observation, weather forecasting, and maritime connectivity.

Q22.

What is confidential VM in Azure?

Answer:

Azure Confidential VMs use AMD SEV-SNP or Intel TDX technology to encrypt VM memory and CPU state, preventing access even from Azure operators. They provide **hardware-rooted attestation** to verify VM integrity before sensitive data is loaded, used in healthcare, finance, and government.

Q23.

What is Azure API Management Developer Portal?

Answer:

The APIM Developer Portal is a customizable website where API consumers can discover APIs, view documentation, obtain subscription keys, test APIs interactively, and manage their subscriptions. It supports custom branding and can be self-hosted, accelerating API adoption by providing a professional API catalog.

Q24.

What is Azure Spot Instance eviction policy?

Answer:

Spot VMs can be evicted when Azure needs capacity, with 30 seconds notice via Scheduled Events. The eviction policy can be set to **Deallocate** (VM stopped, disk retained) or **Delete** (VM and disks deleted). Choose based on whether you need to preserve state; Spot VMs are not suitable for SLA-bound workloads.

Q25.

What is Azure HPC Cache?

Answer:

Azure HPC Cache accelerates NFS-based workloads by aggregating data from multiple storage sources (Azure Blob, NFS NAS) into a single namespace. It provides low-latency access for HPC applications in genomics, oil and gas, and media rendering, eliminating the need to copy large datasets to Azure.

Q26.

What is Azure Elastic SAN backup?

Answer:

Azure Elastic SAN supports integration with Azure Backup for snapshot-based data protection. Snapshots of Elastic SAN volumes are managed through Azure Backup policies, providing consistent backup and recovery for SAN-attached workloads, with application-consistent snapshots using VSS (Windows) or fsfreeze (Linux).

Q27.

What is Azure VMware Solution networking?

Answer:

Azure VMware Solution (AVS) integrates with Azure networking via **ExpressRoute** for connectivity between the private cloud and Azure VNet. AVS uses **NSX-T** for software-defined networking within the private cloud. Traffic can be inspected using **Azure Firewall** or third-party NVAs in the hub VNet.

Q28.

What is Azure Logic Apps Standard VNet integration?

Answer:

Azure Logic Apps Standard supports **VNet integration** (outbound traffic routing through VNet) and **private endpoints** (inbound access restricted to VNet). This enables Logic Apps to securely connect to resources within a VNet (SQL servers, APIs, storage) without exposing them to the public internet.

Q29.

What is Azure Spring Apps and how does it relate to AKS?

Answer:

Azure Spring Apps is built on top of AKS but abstracts away all Kubernetes management. It automatically provisions and manages the underlying AKS cluster, service discovery, configuration server, and distributed tracing. Teams get Spring Boot benefits without needing Kubernetes expertise, trading flexibility for reduced operational overhead.

Q30.

What is Azure Managed Grafana?

Answer:

Azure Managed Grafana is a fully managed Grafana service integrated with Azure Monitor and other Azure data sources. It provides **Azure AD authentication**, **private networking**, **high availability**, and pre-built dashboards for Azure services. It enables building rich, interactive operational dashboards without managing Grafana infrastructure.

Extended Q&A; Bank — Networking (Part 2)

Q1.

What is Azure Private Link?

Answer:

Azure Private Link enables accessing Azure PaaS services and partner services over a private endpoint in your VNet. Unlike Service Endpoints, Private Link traffic flows entirely within the Azure backbone. It also enables exposing your own services privately using a **Private Link Service** for consumers in other VNets.

Q2.

What is Azure Virtual Network Manager?

Answer:

Azure Virtual Network Manager (AVNM) provides centralized management of VNet connectivity and security policies at scale. It creates **network groups**, applies **connectivity configurations** (hub-spoke, mesh), and enforces **security admin rules** that override NSG rules, providing top-down network governance across subscriptions.

Q3.

What is Azure Network Watcher?

Answer:

Network Watcher provides monitoring and diagnostic tools for Azure networking including: **IP Flow Verify** (checks if traffic is allowed/denied by NSG), **Connection Troubleshoot**, **NSG Flow Logs**, **Packet Capture**, **Topology** view, and **Connection Monitor** for end-to-end latency monitoring.

Q4.

What is the difference between Azure Front Door and Azure CDN?

Answer:

Azure Front Door provides Layer 7 load balancing, WAF, SSL offload, URL routing, and session affinity for dynamic content with global anycast routing. **Azure CDN** is optimized for static content delivery with edge caching. Front Door suits dynamic applications; CDN suits static assets. They can be used together.

Q5.

What is BGP in Azure VPN and ExpressRoute?

Answer:

BGP (Border Gateway Protocol) is used for dynamic route exchange in Azure. In **VPN Gateway**, BGP enables dynamic route propagation between on-premises and Azure. In **ExpressRoute**, BGP is mandatory — public and private peering use BGP to advertise Azure and on-premises routes over the dedicated circuit.

Q6.

What is the difference between Service Endpoint and Private Endpoint?

Answer:

Service Endpoints extend VNet identity to Azure services via the Azure backbone but use public IP addresses. **Private Endpoints** assign a private IP from the VNet to the service, making it fully private and accessible from on-premises. Private Endpoints are the recommended approach for secure access to PaaS services.

Q7.

What is an NVA in Azure?

Answer:

A Network Virtual Appliance (NVA) is a VM running a network function such as a firewall, WAN optimizer, or IDS/IPS. Popular NVAs include Palo Alto, Fortinet, and Check Point. NVAs are deployed in a hub VNet and traffic is routed through them via UDRs. Azure Firewall is Microsoft's managed alternative.

Q8.

What is Azure Route Server?

Answer:

Azure Route Server simplifies dynamic routing between NVAs and Azure VNets using BGP. Instead of manually managing UDRs, NVAs exchange routes with Azure Route Server, which propagates them to all VNet-peered connections, enabling dynamic network topologies without manually maintaining route tables.

Q9.

What is forced tunneling in Azure?

Answer:

Forced tunneling redirects all internet-bound traffic from Azure VMs through a VPN Gateway to on-premises for inspection and policy enforcement. Implemented using a **default route (0.0.0.0/0)** in a UDR pointing to the VPN Gateway. Required for compliance scenarios where internet traffic must flow through a corporate security proxy.

Q10.

What is ExpressRoute FastPath?

Answer:

ExpressRoute FastPath improves network performance by bypassing the ExpressRoute Gateway for data path traffic, sending packets directly from on-premises to Azure VMs. This reduces latency and increases throughput. FastPath requires an UltraPerformance or ErGW3AZ gateway SKU.

Q11.

What is connection draining in Azure Application Gateway?

Answer:

Connection draining ensures in-flight requests are completed before a backend instance is removed. When a VM is removed from the backend pool, existing connections complete until the drain timeout expires, preventing request failures during scale-in operations or maintenance deployments.

Q12.

What is Azure Firewall Policy hierarchy?

Answer:

Azure Firewall Policy supports a **parent-child hierarchy**. A base policy defines organization-wide rules; child policies for individual firewalls inherit and extend them. This enables centralized governance via parent policy while allowing local customization through child policies for specific environments.

Q13.

What is Azure Application Gateway WAF?

Answer:

Application Gateway WAF protects web applications using OWASP CRS rule sets. It detects and blocks **SQL injection**, **XSS**, **command injection**, and other OWASP Top 10 threats. WAF Policy can be shared across multiple Application Gateways and supports custom rules and per-URI exclusions.

Q14.

What is ExpressRoute Global Reach?

Answer:

ExpressRoute Global Reach connects on-premises networks to each other through Microsoft's backbone network via their respective ExpressRoute circuits. This allows two data centers in different countries to communicate privately through Azure without traversing the public internet.

Q15.

What is Azure VNet Peering transitivity limitation?

Answer:

VNet Peering in Azure is **non-transitive**. If VNet A peers with VNet B, and VNet B peers with VNet C, VNet A cannot communicate with VNet C through B without a direct peering or hub routing via NVA/Firewall. Azure Virtual WAN or hub-spoke with Azure Firewall solves the transitivity limitation.

Q16.

What is Azure Firewall DNAT?

Answer:

Azure Firewall DNAT translates and filters inbound internet traffic to private Azure resources. A DNAT rule maps a public IP and port on the firewall to a private IP and port of a backend resource, exposing specific private services to the internet without assigning public IPs directly to VMs.

Q17.

What is split-tunnel VPN vs full-tunnel VPN?

Answer:

In **full-tunnel VPN**, all client traffic routes through the VPN, enabling centralized inspection but increasing latency. In **split-tunnel VPN**, only Azure/on-premises traffic routes through the VPN; internet traffic goes directly. Split tunneling reduces VPN gateway load and improves remote worker performance.

Q18.

What is Azure Virtual Network Encryption?

Answer:

Azure VNet Encryption encrypts traffic between VMs within the same VNet or peered VNets using DTLS. It provides encryption for VM-to-VM communication without modifying applications, addressing compliance requirements for encryption of data in transit within the network perimeter. Requires supported VM SKUs.

Q19.

What is Azure Private DNS Resolver inbound endpoint?

Answer:

The Private DNS Resolver inbound endpoint provides a private IP within your VNet that on-premises DNS resolvers can forward queries to. When on-premises systems forward queries for Azure Private DNS zone names to this IP, the resolver resolves them against Azure Private DNS zones, enabling split-horizon DNS for hybrid environments.

Q20.

What is NAT Gateway and when is it preferred over Load Balancer SNAT?

Answer:

Azure NAT Gateway is preferred over Load Balancer SNAT when you need **predictable outbound connectivity**, need to avoid SNAT port exhaustion (NAT Gateway provides up to 64,512 ports per public IP), or when VMs in private subnets need outbound internet access without individual public IPs.

Extended Q&A; Bank – Security & Monitoring (Part 2)

Q1.

What is Azure Security Score?

Answer:

Secure Score in Microsoft Defender for Cloud measures the security posture of your Azure environment as a percentage. Improving the score by implementing recommendations reduces attack surface. It provides a prioritized remediation plan aligned to MCSB and regulatory standards.

Q2.

What is Azure Monitor Metrics vs Logs?

Answer:

Metrics are numerical, time-series values collected at frequent intervals, ideal for alerting and dashboarding with low latency. **Logs** are structured or unstructured records in Log Analytics Workspace, queried with KQL, ideal for troubleshooting, auditing, and complex correlation across multiple data sources.

Q3.

What is Microsoft Defender for DevOps?

Answer:

Microsoft Defender for DevOps provides security management across DevOps pipelines in GitHub, Azure DevOps, and GitLab. It performs **infrastructure-as-code scanning**, **secret scanning**, **code scanning**, and **container image scanning** during CI/CD, exposing findings in Defender for Cloud with pull request annotations.

Q4.

What is the difference between diagnostic settings and audit logs?

Answer:

Diagnostic settings collect resource-level metrics and logs and route them to Log Analytics, Storage, or Event Hub. **Azure Activity Log** specifically records all subscription-level management operations (who did what and when), available for 90 days and exportable via diagnostic settings.

Q5.

What is Microsoft Sentinel Analytics Rules?

Answer:

Sentinel Analytics Rules define conditions generating security alerts and incidents. Types: **Scheduled** (KQL queries on a schedule), **NRT (Near Real-Time)**, **Microsoft Security** (alerts from Defender products), **Fusion** (ML-based multi-signal correlation), and **Anomaly** (ML-based behavioral detection).

Q6.

What is an Azure Monitor Action Group?

Answer:

An Action Group defines notifications and actions triggered when an alert fires. Actions include **email/SMS/push notifications**, **voice calls**, **Azure Functions**, **Logic Apps**, **ITSM connectors** (ServiceNow, Jira), **runbooks**, and **webhooks**. Multiple alerts can share the same action group.

Q7.

What is change analysis in Azure Monitor?

Answer:

Azure Monitor Change Analysis tracks changes across Azure subscriptions including **resource property changes** (ARM-level), **application setting changes**, and **web app in-guest changes**. It helps correlate application issues with recent infrastructure or configuration changes, reducing MTTR.

Q8.

What is the principle of defense in depth?

Answer:

Defense in depth uses multiple layers of security controls so that if one fails, others still protect the system. Azure layers include: **physical** (Microsoft DCs), **perimeter** (DDoS, Firewall), **network** (NSGs), **compute** (Defender for Servers), **application** (WAF), **data** (encryption), and **identity** (MFA, PIM).

Q9.

What is Azure Monitor Private Link Scope (AMPLS)?

Answer:

AMPLS connects Azure Monitor resources (Log Analytics Workspaces, Application Insights) to your VNet using Private Endpoints. This ensures monitoring data travels over the Azure backbone, and on-premises agents send data to Azure Monitor without internet access. AMPLS enforces which workspaces accept data from private networks only.

Q10.

What is Defender for Cloud Governance?

Answer:

Defender for Cloud Governance assigns owners and due dates to security recommendations, tracking remediation accountability. It generates **governance reports** showing assigned, remediated, or overdue recommendations and integrates with Azure DevOps and Jira to create work items from security recommendations.

Q11.

What is Microsoft Sentinel Playbooks?

Answer:

Sentinel Playbooks are Logic Apps triggered by security alerts or incidents for automated response. They can **enrich incidents** with threat intelligence, **notify teams** via email or Teams, **block IPs** in NSGs, **disable compromised accounts** in Azure AD, or **create ITSM tickets**, enabling SOAR capabilities.

Q12.

What is the difference between SIEM and SOAR?

Answer:

SIEM collects, correlates, and analyzes security events, providing detection and investigation capabilities. **SOAR** automates response to detected threats via playbooks and workflows. Microsoft Sentinel combines both capabilities in a single cloud-native platform for unified security operations.

Q13.

What is Defender for APIs?

Answer:

Microsoft Defender for APIs provides security posture management and threat detection for APIs published through Azure API Management. It discovers API endpoints, classifies sensitive data exposure, and detects anomalous patterns like **parameter tampering**, **data exfiltration**, and **unusual caller behavior**.

Q14.

What is Azure Trusted Launch?

Answer:

Azure Trusted Launch for Generation 2 VMs provides **Secure Boot** (prevents malicious bootloader injection), **vTPM** (enables secure boot measurements and BitLocker), and **integrity monitoring**. It protects against rootkits, bootkits, and kernel-level malware through hardware-rooted security guarantees.

Q15.

What is the difference between Azure Firewall Standard and Premium?

Answer:

Azure Firewall Standard provides stateful L4/L7 inspection, FQDN filtering, and threat intelligence. **Azure Firewall Premium** adds **TLS inspection** (decrypts and inspects HTTPS traffic), **IDPS** (58,000+ signatures), and **URL filtering**. Premium is required for advanced threat protection and compliance.

Q16.

What is Azure Monitor Workbooks?

Answer:

Azure Monitor Workbooks are interactive reports combining text, KQL queries, metrics, parameters, and rich visualizations. They support **custom templates**, **parameterized queries**, and **cross-resource queries**, widely used for security dashboards, compliance reporting, and operational runbooks.

Q17.

What is Azure Key Vault Firewall?

Answer:

Azure Key Vault Firewall restricts access to specific virtual networks and IP addresses. Combined with Private Endpoints, it ensures Key Vault is only accessible from trusted networks. The 'Allow trusted Microsoft services' option must be enabled for Azure Backup and Azure Monitor to continue accessing Key Vault.

Q18.

What is Azure role assignment condition?

Answer:

RBAC role assignment conditions allow fine-grained access control based on resource attributes. For example, allowing a user to read only blobs with a specific tag value, or restricting access to storage accounts in specific regions. Conditions reduce the need for overly broad permissions using attribute-based access control (ABAC).

Q19.

What is Azure Monitor Insights?

Answer:

Azure Monitor Insights are curated monitoring experiences for specific Azure services with pre-built workbooks, dashboards, and alerts. Examples: **VM Insights**, **Container Insights** (AKS monitoring), **Network Insights**, **Storage Insights**, and **SQL Insights**. They reduce effort of building monitoring from scratch.

Q20.

What is the Microsoft Security Response Center (MSRC)?

Answer:

MSRC investigates reports of security vulnerabilities in Microsoft products and services. It manages Microsoft's responsible disclosure program, coordinates with security researchers, and publishes security advisories. In Azure, MSRC coordinates patch and disclosure of vulnerabilities via monthly Patch Tuesday releases.

Extended Q&A; Bank – Business Continuity (Part 2)

Q1.

What is the Azure paired region strategy?

Answer:

Azure pairs regions within the same geography. Paired regions receive updates sequentially, reducing risk. In disasters, Microsoft prioritizes recovery of one region in each pair. Customers should align their DR strategy with Azure's pairing to leverage Microsoft's priority recovery commitments.

Q2.

What is ASR replication policy?

Answer:

An ASR replication policy defines: **recovery point retention** (how long recovery points are stored), **app-consistent snapshot frequency** (VSS snapshots for application consistency), and **crash-consistent recovery point frequency**. For Azure-to-Azure, crash-consistent points are every 5 minutes; app-consistent points are configurable from 1-12 hours.

Q3.

What is multi-region active-active vs active-passive?

Answer:

Active-active: Both regions simultaneously serve traffic with near-zero RTO and real-time data sync. Higher cost. **Active-passive:** Secondary region on standby with lower cost but higher RTO. Cosmos DB multi-region write enables active-active for globally distributed applications with automatic failover.

Q4.

What is RPO for Azure SQL Database automated backups?

Answer:

Azure SQL Database performs **full backups** weekly, **differential backups** every 12-24 hours, and **transaction log backups** every 5-12 minutes, providing an effective RPO of approximately 5-12 minutes for point-in-time restore within the 1-35 day configurable retention period.

Q5.

What is Azure Backup for AKS?

Answer:

Azure Backup for AKS enables backup and restoration of AKS cluster configurations and persistent volume data using a **Backup Extension** installed in the cluster. It captures Kubernetes resources and CSI snapshots, stored in a Backup Vault, restorable to the same or different AKS cluster.

Q6.

What is geo-zone-redundant storage (GZRS)?

Answer:

GZRS combines ZRS and geo-redundancy. Data is replicated synchronously across **three availability zones** in the primary region and asynchronously to the paired region. It protects against zone-level failures and regional disasters. RA-GZRS additionally allows reading from the secondary region.

Q7.

What is Recovery Services Vault vs Backup Vault?

Answer:

Recovery Services Vault supports Azure VM backup, SQL/SAP HANA in Azure VMs, Azure Files, and Azure Site Recovery. **Backup Vault** is a newer resource supporting Azure Disks backup, Azure Blobs backup, Azure Database for PostgreSQL, and AKS backup. Microsoft is gradually migrating workloads to Backup Vault architecture.

Q8.

What is immutable vault in Azure Backup?

Answer:

Immutable vault prevents deletion of backup data and changes to backup policies that reduce retention. Operations like **stopping backup with delete data** and **reducing retention period** are blocked. Immutable vault can be made **irreversible (locked)**, preventing even subscription administrators from disabling it.

Q9.

How do you test DR plans in Azure without impacting production?

Answer:

Test DR using: **ASR test failover** (spins up a replica in an isolated network without affecting production), **Azure Backup restore to alternate resource**, **Azure Chaos Studio** (inject faults), and **runbook testing**. Conduct DR drills quarterly for critical workloads and document RTO/RPO achieved.

Q10.

What is cross-subscription restore in Azure Backup?

Answer:

Cross-subscription restore (CSR) enables restoring Azure VM backups to a different subscription than the source. Useful for DR scenarios where the production subscription may be compromised or unavailable. Supported for Azure VMs and managed disks when using Recovery Services Vault with appropriate permissions.

Q11.

What is Azure Front Door health probes for business continuity?

Answer:

Azure Front Door uses HTTP/HTTPS health probes sent to backend endpoints to determine availability. Based on health probe responses, Front Door automatically routes traffic away from unhealthy backends. Health probes are configured per **origin group** with customizable probe interval, probe path, and probe method.

Q12.

What is the difference between hot, warm, and cold standby?

Answer:

Hot standby is a fully operational replica with real-time data sync and near-instantaneous failover — highest cost. **Warm standby** is a scaled-down but running environment with regular data sync — moderate cost and RTO. **Cold standby** requires rebuilding from backups — lowest cost, highest RTO.

Q13.

What is soft delete for Recovery Services Vault?

Answer:

Soft delete for Azure Backup retains deleted backup data for 14 additional days at no extra cost. Enhanced soft delete extends retention up to 180 days and can be made irreversible, providing ransomware protection by preventing malicious actors from deleting backup data even if they gain access to the subscription.

Q14.

What is Azure Elastic SAN and business continuity?

Answer:

Azure Elastic SAN supports zone-redundant deployment for high availability against zone failures and integrates with Azure Backup for snapshot-based recovery. For DR, Elastic SAN volumes can be snapshotted and replicated to another region using managed disk infrastructure, providing RTO/RPO aligned to snapshot frequency.

Q15.

What is the Azure Business Continuity Center unified view?

Answer:

The Azure Business Continuity Center consolidates **Azure Backup** and **Azure Site Recovery** management across all protected workloads. It provides centralized policy management, compliance reporting, alert management, and a single operational view of protection status across the enterprise from one portal experience.
